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White Rabbit Switch (WRSv4)

Ultra-Precise Time Synchronization & Deterministic Ethernet Solution Qbit Labs’ White Rabbit Switch (WRSv4) delivers subnanosecond time synchronization and deterministic Ethernet switching for missioncritical applications. Based on CERNled White Rabbit technology, the WRSv4 combines a highprecision boundary clock with a 24port Gigabit Ethernet Layer-2 switch, enabling phase-aligned, frequencylocked, and timeaccurate data delivery over standard optical fiber up to 100 km. Built on open hardware and software, the WRSv4 is ideal for scientific facilities, telecom networks, smart grids, finance, and defense systems where absolute time accuracy, low jitter, and deterministic latency are mandatory.

Introduction

The Qbit White Rabbit Switch v4 (WRSv4) is a high-precision Ethernet switch designed to provide deterministic data transfer along with sub-nanosecond time synchronization over standard optical fiber networks. Developed within the White Rabbit collaboration led by CERN, it combines advanced timing technology with reliable Layer-2 switching capabilities.

The WRSv4 integrates a 24-port Gigabit Ethernet switching fabric with an enhanced boundary clock that extends IEEE 1588-2008 (PTP) to deliver accurate phase, frequency, and absolute time distribution. By using Synchronous Ethernet, hardware timestamping, and precise clock alignment techniques, it ensures highly stable and low-latency network performance.

Built on open-source hardware and software, the switch supports features such as QoS control, VLAN processing, monitoring tools, and robust management interfaces. It is optimized for mission-critical applications including large scientific facilities, telecommunications, smart grids, finance, and defense systems that require ultra-precise timing and reliable network operation.

Architecture

The White Rabbit Switch (WRSv4) architecture is built around an FPGA-based switching core combined with multiple White Rabbit–enabled Gigabit Ethernet ports to provide deterministic data transfer and sub-nanosecond time synchronization.

It includes a dedicated White Rabbit timing engine that implements SyncE and IEEE 1588/PTP mechanisms for precise clock recovery, alignment, and distribution across all ports. A low-latency, non-blocking switching fabric handles packet forwarding, VLAN processing, QoS enforcement, and buffering for reliable real-time communication.

The system also integrates high-stability clock generation, hardware timestamping, and an embedded management subsystem for configuration, monitoring, diagnostics, and secure operation, along with robust power and environmental monitoring for reliable performance in demanding environments.

Why Choose Us?

🚀 Rapid integration, faster time-to-market 📘 Clean, verified, documented RTL 🤝 Direct access to experts

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Features

- **Switching**

The WRSv4 features a 24-port Gigabit Ethernet Layer-2 switching fabric based on SFP optical interfaces, enabling flexible fiber connectivity across distributed networks. It supports deterministic, low-latency packet forwarding to ensure predictable data delivery for time-critical applications, while remaining fully compatible with standard Ethernet infrastructure and protocols.

- **Time Synchronization (White Rabbit PTP)**

The switch provides ultra-precise time synchronization using White Rabbit extensions to IEEE 1588-2008 (PTPv2). It achieves sub-nanosecond accuracy (typically <500 ps, guaranteed <1 ns) through hardware timestamping, Synchronous Ethernet, and continuous calibration of fiber delay and link asymmetry. It also supports external timing references via 1PPS and 10 MHz input/output signals.

- **Grandmaster & Boundary Clock Modes**

WRSv4 can operate as a GNSS-disciplined Grandmaster clock for distributing highly accurate network time or function as a Boundary Clock (Simple Master) in cascaded or daisy-chained timing architectures. It also supports high-precision event time-tagging with picosecond-level resolution.

- **Reliability & Availability**

Designed for mission-critical deployments, the switch supports redundant power supplies and incorporates high-stability oscillators for holdover operation during timing reference loss. Its hardware is built for continuous 24×7 operation in demanding environments.

- **Management & Control**

The system includes secure remote management via SSH and SNMP, along with a web-based interface for configuration, monitoring, and diagnostics. Firmware and FPGA gateware can be upgraded in the field, ensuring long-term maintainability and feature updates.

Specification

SYNCHRONIZATION & PERFORMANCE	
Parameter	**Specification**
Time Synchronization Accuracy	• < 500 ps typical, < 1 ns guaranteed
Jitter Performance	• Picosecond-level
Data Delivery	• Fully deterministic
Maximum Fiber Distance	• Up to 100 km
Packet Forwarding	• Deterministic low-latency switching
Fiber Calibration	• Dynamic asymmetry calibration
SWITCH & NETWORK INTERFACES	
Total Ports	• 24 × SFP+ ports
Active Ports	• 20 × 1 Gb SFP
Reserved Ports	• 4 × SFP ports

Specification Cont...

Management SFP Port	• 1 × 1Gb SFP
RJ45 Ethernet Port	• 1 × 10/100/1000 Mbps
UART	• RS-232 over RJ45
USB Interfaces	• USB-C (UART), USB-A
CLOCKING INTERFACES (SMA)	
PPS Input	• LVCMOS-3.3V / 5V TTL compatible, selectable Hi-Z or 50Ω termination
10 MHz Input	• Sine or digital, −10 dBm to +24 dBm
PPS Output	• TTL-5V standard PPS
10 MHz Output	• AC-coupled 10 MHz
Auxiliary Port	• AUX (abscal)
PROGRAMMABLE PROCESSING RESOURCES	
SoC / FPGA	• Xilinx Zynq Ultrascale+ XCZU17EG
Application CPU	• Quad-core ARM Cortex-A53 up to 1.5 GHz
Real-time CPU	• Dual-core ARM Cortex-R5 up to 600 MHz
High-speed Transceivers	• 32 GTH (16.3 Gb/s), 16 GTY (32.75 Gb/s)
CLOCKING RESOURCES	
Main Oscillator	• 25 MHz VCTCXO with 16-bit DAC control
DAC	• AD5662 16-bit
Auxiliary Oscillators	• 2 × Si549
Grandmaster PLL	• LMX2594
Main Loop PLL	• HMC7044

Specification Cont...

MEMORY & STORAGE	
System Memory	4 GB SO-DIMM
eMMC Storage	4 GB
NOR Flash	Dual Quad-SPI (2 × 1 Gb)
EEPROM	I²C Serial EEPROM 64K
Expansion Storage	M.2 SATA SSD support
TIMING & SYNCHRONIZATION	
Timing Protocol	IEEE 1588-2008 PTP with White Rabbit extensions
Grandmaster Mode	Supported
Boundary Clock Mode	Supported
PPS Distribution	Supported
Reference Clock I/O	10 MHz / 62.5 MHz
Event Timestamping	Hardware-based precise tagging
NETWORKING & PROTOCOL SUPPORT	
Switching Standards	IEEE 802.1x
VLAN	IEEE 802.1Q
Multicast	IGMP snooping
Loop Prevention	STP / RSTP
IP Stack	TCP/IP, DHCP, ARP, DNS, TFTP
Remote Access	SSH, SNMP v1/v2c
Time Backup	NTP support
MANAGEMENT & CONFIGURATION	
Web Interface	Browser-based management

Specification Cont...

CLI Access	RS-232 / USB UART
SNMP Monitoring	Supported
Firmware Upgrade	Field upgradable
USER INTERFACE & INDICATORS	
Display	OLED
Status LEDs	Programmable red/green
Buttons	Reset + 2 control buttons
Rear Panel Indicator	Location LED
MECHANICAL SPECIFICATIONS	
Form Factor	1U Rack Mount
Height	44.45 mm
Width	482.6 mm
Depth	310 mm
ENVIRONMENTAL SPECIFICATIONS	
Operating Temperature	0°C to +45°C
Storage Temperature	−25°C to +60°C
Humidity	10% to 90% non-condensing

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